

Silin Raj Gupta

📞 519-505-7421 ✉ sr4gupta@uwaterloo.ca 🔗 LinkedIn 🐙 GitHub 📁 Portfolio

EDUCATION

University of Waterloo

Bachelor of Computer Science, Honours, Co-op

Sep 2024 – Present

SKILLS

Languages: Python, Java, C, C++, C#, JavaScript/TypeScript, Racket, Lisp, Scheme, HTML/CSS, SQL

Frameworks: React.js, Next.js, Node.js, Django, Flask, FastAPI, GraphQL, Tailwind CSS

ML/AI: TensorFlow, Scikit-learn, Pandas, NumPy, LangChain, RAG, Vector Embeddings

APIs/Integrations: OpenAI, Google Gemini, Grok, Sonar (Perplexity) Spotify, Twilio, ElevenLabs, VAPI

Tools: Git, Bash, Docker, Pinecone, Vercel, Render, AWS (EC2), CI/CD, Microsoft Azure, Postman, Figma

PROFESSIONAL EXPERIENCE

e3 Group

Full-Stack AI Engineer

- Incoming Winter 2026

Jan 2026 – Apr 2026 | San Francisco, CA

Merazine

Software Engineer Intern

- Developed a full-stack AI companion mobile app (React Native, Node.js, GraphQL) featuring real-time voice conversations powered by OpenAI GPT-4o, Whisper STT, and ElevenLabs TTS, with long-term memory using vector embeddings for personalized user interactions.
- Built digital-physical product integration connecting a physical AI doll to the mobile app, enabling users to customize the doll's wardrobe through an interactive avatar system with real-time preview and cloud synchronization.
- Architected production infrastructure including centralized error handling, configurable rate limiting, TTL-based caching, and a cost tracking system with admin dashboard to monitor API spend across OpenAI, ElevenLabs, and mem0.
- Designed conversational AI system with short-term memory (conversation context) and long-term memory (user preferences, facts) to create natural, context-aware responses that evolve with each user over time.

Oct 2025 – Dec 2025 | San Francisco, CA

PolicyAdvisor

Software Engineer Intern

- Developed, tested, and debugged a Next.js + Python (FastAPI) proposal-automation platform powered by Google Gemini API; accelerated processing from 5–6 hrs → 2–5 mins with 99%+ accuracy across 10+ carriers, driving ~\$100,000 in annual operational savings.
- Spearheaded a visitor insurance sales platform (vIQ) by leading a cross-functional team; launched in 2 weeks, integrating 10+ carrier APIs and intelligent caching.
- Architected an internal life-insurance eligibility chatbot (LLM + carrier rules) using Next.js/TypeScript with Pinecone-backed RAG and vector embeddings, streamlining advisor workflows during live calls.
- Orchestrated a 24/7 AI voicebot (n8n + ElevenLabs + Twilio + Google Calendar) that handled client calls and auto-scheduled callbacks, reducing missed leads by 40%.
- Designed an automated blog QA pipeline using Perplexity, Grok, and OpenAI APIs across 5 checkpoints, generating polished HTML review reports.

May 2025 – Aug 2025 | Toronto, ON

PROJECTS

SpeakSummarize - Record. Recall

- Built AI speech-to-notes app with structured summaries, action items, and multi-speaker diarization.
- Reached 1,000+ downloads and around \$1K MRR within days of launch.
- Used Whisper-1 for robust speech-to-text, GPT-4o-mini for fast low-cost summaries, and Pinecone for semantic search over transcripts.
- Added Mem0-based memory to learn user preferences and drove initial growth via Reddit; now extending to Android and watchOS.

Muse Sketch Studio [Replicate AI Hackathon Winner]

- Built a complete AI-powered fashion design workflow transforming text prompts into professional runway videos through 5-step pipeline (sketch → colors → model → 3D → runway).
- Integrated Google nano-banana and Kling v2.1 models via Replicate API for professional fashion sketches and video generation.
- Developed responsive React/TypeScript frontend with real-time generation tracking and downloadable outputs.

Moosic - AI Playlist Generator

- Launched a website generating 100+ playlists weekly by combining Spotify and OpenAI APIs for playlist creation.
- Built seamless Spotify API integration for authentication, playlist creation, and top song retrieval.
- Deployed a scalable architecture (Render backend + Vercel frontend) with 100% uptime.